1. Project Title

AI-Powered Prescription Recognition and Intelligent Medication Reminder System

2. Problem Statement

In Bangladesh, most prescriptions are handwritten, and use shorthand dose patterns such as "1+0+1" (morning, noon, night). Many patients, especially elderly people, find these prescriptions hard to read and hard to follow. As a result, doses are missed, medicines are taken at the wrong time, and patients do not fully understand what they are taking or why.

Existing solutions each solve only part of this problem. Global reminder apps such as Medisafe require the user to enter every medicine manually and are not built for Bangladeshi prescriptions. Local apps such as DigiCure BD and Daktarbhai offer reminders and e-prescriptions but cannot read a paper prescription. Medicine index apps such as MedEx Plus give medicine details and prices but have no reminders and no prescription scanning. Research prototypes can read prescription text, but they stop at text extraction, report limited or no real-world accuracy, and are not usable by ordinary patients.

There is no accessible tool that takes a photo of a handwritten Bangladeshi prescription and turns it directly into a reliable, confirmed reminder schedule.

Target Users: Everyone from elderly individuals and their caregivers to busy professionals managing their health without full-time support.

3. Objectives

- Build a mobile application that reads a photo of a handwritten prescription and extracts medicine names, doses, durations and timing patterns as structured data.
- Let the user review, correct and confirm the extracted schedule before it is saved, so that no unverified reading is trusted automatically.
- Generate and send timely medicine reminders, and track whether each dose was taken or missed.
- Provide plain-language information about each medicine, including its generic name and general price information, to help users understand their treatment.
- Gently flag a medicine that does not seem to match the stated condition, and advise the user to consult a doctor or pharmacist.
- Support elderly patients and their caregivers through a simple, responsive interface that works on mobile phones.
- Deliver a complete, deployed, documented system following the course workflow.

4. Features

- User registration and login with secure authentication.
- Prescription photo upload and image storage.
- AI-based extraction of medicine name, dose, duration and timing from the photo.
- Confirm-and-edit screen for the extracted schedule, with low-confidence readings clearly highlighted.

- Automatic reminder schedule creation from the confirmed prescription.
- Scheduled reminders for each dose.
- Dose tracking (taken, missed, skipped) with history.
- Medicine information page with a plain-language explanation, generic name and general price data.
- Condition-based caution note advising the user to consult a doctor or pharmacist when a medicine seems unrelated to the stated condition.
- Dashboard showing today's doses, adherence summary and active prescriptions.
- Caregiver view so a family member can monitor a patient's adherence (if time permits).
- Responsive design that works on mobile.

5. Technology Stack

- **Frontend:** React (Vite) with Tailwind CSS, Capacitor shell.
- **Native Plugins:** @capacitor/local-notifications , @capacitor/camera, @capacitor/filesystem, @capacitor/push-notifications.
- **Backend:** Node.js with Express, JSON Web Token (JWT) authentication, bcrypt password hashing, node-cron.
- **Database:** PostgreSQL (hosted on the Supabase free plan), Supabase Storage (persistence of prescription images).
- **AI Tools:** Google Gemini API (multimodal).
- **Deployment:** Render (free web service), Vercel/Netlify (free tier), Supabase, GitHub Releases/Firebase App Distribution.
- **Version Control and Tooling:** GitHub for source control, Postman for API testing, Figma for UI/UX design, Android Studio for build compilation and release signing, and Vitest or Jest for automated testing.

6. Expected Outcome

User Benefits: Users will be able to photograph a prescription and obtain a reminder schedule within minutes, needing only to confirm or edit the extracted details rather than entering each medicine and timing manually. Elderly patients and their caregivers will gain a clearer understanding of their treatment, a dependable means of avoiding missed doses, and access to general medicine information and reference pricing.

Workflow Improvement: The existing workflow requires patients to interpret handwriting by eye, set phone alarms manually, and search for medicine details separately. The proposed system consolidates prescription reading, scheduled reminders, dose tracking and medicine information in a single application. The confirm-and-edit step keeps the user in control of the final schedule, and the system is explicit that it supports, and does not replace, the advice of a doctor or pharmacist.

Benefits of AI Integration: The multimodal AI model is expected to interpret handwritten and mixed-language prescriptions, which rule-based methods cannot handle reliably, and to convert them into structured data covering medicine names, doses and timing patterns. It also generates plain-language medication summaries and a cautious, confidence-based condition check, reducing manual effort and the likelihood of data-entry errors. The project will be delivered as a deployed, tested and documented full-stack application with meaningful AI integration, following the course workflow.